

Lab Experience Report: Jupyter and GitHub Exploration

The link to my GitHub repository:

<https://github.com/binxixi23/jupyter-exploration>

Part A: What I Did

In this lab session, I completed the setup for two major tools: GitHub and Jupyter Notebooks. I followed a step-by-step process to connect my local computer to the cloud.

1. **GitHub Account Setup:** I went to the official website and signed up for a free account using my username binxixi23.
2. **Repository Creation:** I created a new public project folder called jupyter-exploration. I checked the box to include an initial README.md text file.
3. **Editing the README:** I opened the README file directly on the website. I edited it to add a short description of our lab goals and saved the changes.
4. **Jupyter Notebook Creation:** I opened my Jupyter environment and created a new file. I added a Markdown cell with titles and text. Then, I added a Python code cell using the `print("Hello, World!")` command and ran both cells using the Shift + Enter shortcut.
5. **Saving and Syncing:** I saved the notebook file as `My_First_Notebook.ipynb`. I downloaded it to my computer, installed the **GitHub Desktop** application, and used it to copy, commit, and push my final notebook file up to the website.

Part B: What I Learned

This lab taught me the foundational concepts of modern software development and cloud storage.

1. **Version Control Basics:** I learned that GitHub acts like a time machine for code. It allows me to save different versions of my work safely in the cloud using repositories, commits, and pushes.

2. **Interactive Computing:** I discovered that Jupyter Notebooks are incredibly useful because they combine normal reading text (Markdown) and live programming code (Python) in the exact same file.
3. **Markdown Text:** I learned that Markdown is just a simple plain text format.

Challenges I Faced and Overcame

My main challenge happened inside the **GitHub Desktop** application. When I added my `My_First_Notebook.ipynb` notebook file, the font looked completely broken and scrambled on my screen. I saw a lot of weird background code, brackets, and quotes.

I was tempted to change or edit the text to fix it, but I decided to leave it alone and clicked "Push" anyway. When I checked the official GitHub website, the notebook rendered beautifully with perfect text and code blocks. I learned that notebook files are stored as raw text data in the background, and I should never edit that text directly inside GitHub Desktop, or the file might break.